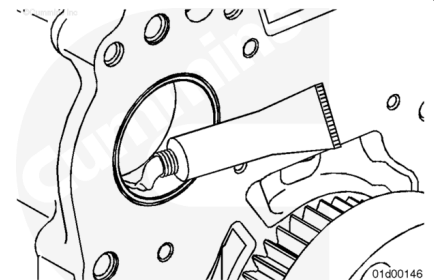


Trainings ▾

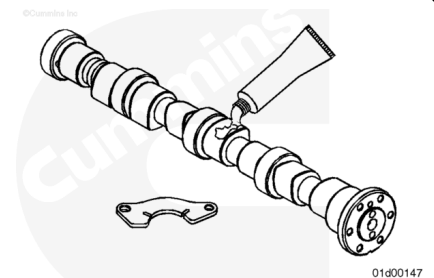
Install

Rear Gear Train

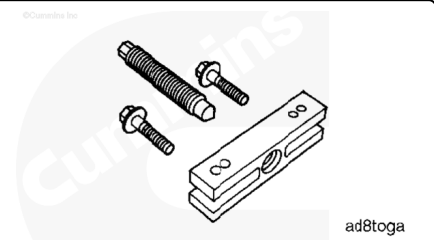
Apply assembly lubricant, Part Number 3163087, to the rear camshaft bore.



Lubricate the camshaft lobes, journals, and thrust washer with assembly lubricant, Part Number 3163087.

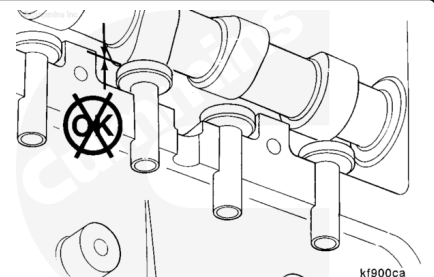


Use a gear puller, service tool Part Number ST647, or equivalent, to attach to the end of the camshaft where the camshaft gear mounts, to act as a handle. This will give proper leverage and ease installing the camshaft.



⚠ CAUTION ⚠

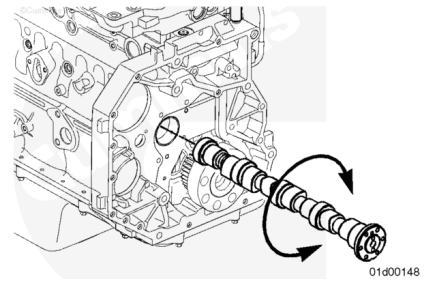
Before the camshaft is installed, verify the tappets are fully inserted into the tappet bores. If the tappets are not fully inserted, the camshaft and/or tappets may be damaged during installation.



Note : The following illustrations will show the engine in the upright position for clarity.

⚠ CAUTION ⚠

Do not force the camshaft into the camshaft bore as damage to the camshaft bushing can result.

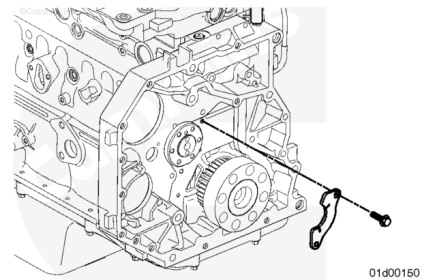


Install the camshaft. While pushing in slightly, rotate the camshaft and carefully work the camshaft through the camshaft bushings. As each camshaft journal passes through a bushing, the camshaft will drop slightly and the camshaft lobes will catch on the bushings. Rotating the camshaft will free the lobe from the bushing and allow the camshaft to be installed.

Install the thrust plate.

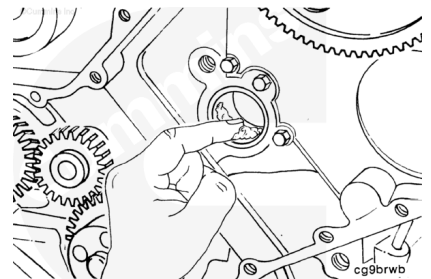
Install the thrust plate capscrews.

Torque Value: 24 n•m [212 in-lb]

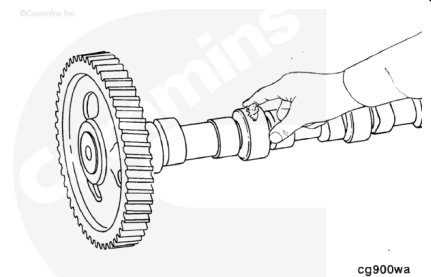


Front Gear Train

Apply assembly lubricant, Part Number 3163087, to the front camshaft bore.



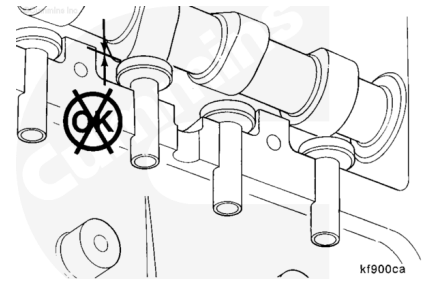
Lubricate the camshaft lobes, journals, and thrust plate with assembly lubricant, Part Number 3163087.



⚠ CAUTION ⚠

Before the camshaft is installed, verify the tappets are fully inserted into the tappet bores. If the tappets are not fully inserted, the camshaft and/or tappets may be damaged

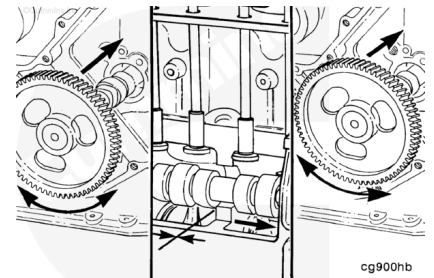
during installation.



Note : The following illustrations will show the engine in the upright position for clarity.

⚠ CAUTION ⚠

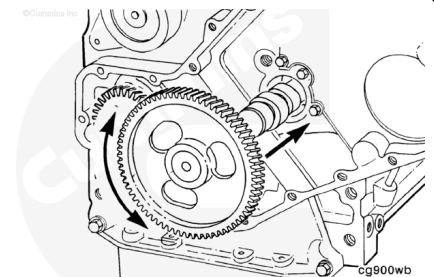
Do not try to force the camshaft into the camshaft bore as damage to the camshaft bushing can result.



Install the camshaft. While pushing in slightly, rotate the camshaft and carefully work the camshaft through the camshaft bushings. As each camshaft journal passes through a bushing, the camshaft will drop slightly and the camshaft lobes will catch on the bushings. Rotating the camshaft will free the lobe from the bushing and allow the camshaft to be installed.

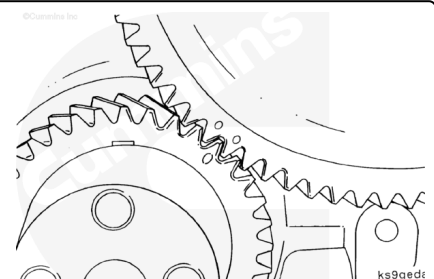
⚠ CAUTION ⚠

To reduce the possibility of engine damage, make sure the camshaft rotates freely.



Before the camshaft gear engages the crankshaft gear, check the camshaft for ease of rotation. When installed properly, the camshaft **must** rotate freely.

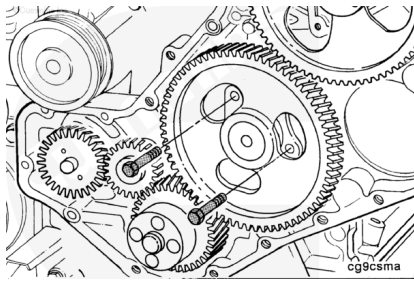
Align the timing marks as illustrated and finish installing the camshaft.



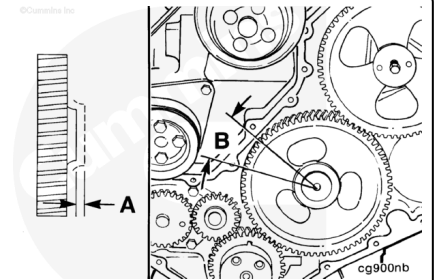
Install the thrust plate.

Install the thrust plate capscrews.

Torque Value: 24 n·m [18 ft·lb]



Use gauge, Part Number 3824564, and magnetic base, Part Number 3377399, to verify the camshaft has proper backlash and end clearance.



Camshaft End Play (A)

mm		in
0.12	MIN	0.005
0.47	MAX	0.018

Camshaft Gear Backlash Limits (B)

mm		in
0.076	MIN	0.003
0.280	MAX	0.011